

pharmaniaga®

AMIKACIN

500MG/2ML

INJECTION



DESCRIPTION

Colourless to pale yellow solution.

The compatible diluents for intravenous use if required are as follows:

- 5% Dextrose in water
- 5% Dextrose and 0.2% Sodium Chloride Injection
- 5% Dextrose and 0.45% Sodium Chloride Injection
- 0.9% Sodium Chloride Injection
- Isotonic Sodium Chloride Injection
- Lactated Ringer's Injection

COMPOSITION

Each 2ml contains Amikacin 500mg.

PHARMACODYNAMICS

Amikacin is a semi-synthetic aminoglycoside antibiotic derived from Kanamycin A. It is active against a broad spectrum of Gram-negative organisms, including *Pseudomonas*, *Escherichia coli* and some Gram-positive organisms, e.g. *Staphylococcus aureus*.

Aminoglycoside antibiotics are bactericidal in action. Although the exact mechanism of action has not been fully elucidated, the drugs appear to inhibit protein synthesis in susceptible bacteria by irreversibly binding to 30S ribosomal subunits.

PHARMACOKINETICS

Amikacin is rapidly absorbed after intramuscular (IM) injection. Peak serum levels of approximately 11 mcg/ml and 20 mcg/ml are reached one hour after IM administration of dose of 250 mg and 500 mg respectively. Levels 10 hours after injection are on the order of 0.3 mcg/ml and 2.1 mcg/ml respectively. Serum concentrations are detectable at 12 hours. Amikacin is negligibly bound to serum protein. Amikacin diffuses readily through extracellular fluids and is excreted in the urine unchanged primarily by glomerular filtration. Half-life in individuals with normal renal function is 2-3 hours.

Following intramuscular administration of 250 mg dose, about 65% is excreted in six hours and 91% within 24 hours. The urinary concentrations average 563 mcg/ml in the first six hours and 163 mcg/ml over 6 to 12 hours. Mean urine concentrations after a 500 mg IM dose average 832 mcg/ml.

Single dose of 500 mg administered to normal adults as an intravenous infusion over a period of 30 minutes produced a mean peak serum concentration of 38 mcg/ml at the end of the infusion. Repeated infusions did not reveal any drug accumulation.

INDICATION

Pharmaniaga Amikacin 500mg/2ml Injection is indicated in the short-term treatment of serious infections caused by susceptible strains of Gram-negative bacteria.

Staphylococcus aureus, including methicillin resistant strains is the principal Gram-positive organism sensitive to amikacin. The use of amikacin in the treatment of Staphylococcal infections should be restricted to second-line therapy, and should be confined to patients suffering from severe infections caused by susceptible strains of staphylococcus who have failed to respond or are allergic to other available antibiotics.

Pharmaniaga Amikacin 500mg/2ml Injection is indicated in the treatment of neonatal sepsis when sensitivity testing indicates that other aminoglycosides cannot be used. In certain severe infections such as neonatal sepsis, concomitant therapy with a penicillin type drug may be indicated because of the possibility of infections due to Gram-positive organisms such as streptococci or pneumococci. If concomitant treatment with a penicillin type drug is indicated, then the drugs should be administered separately because in-vitro mixing of the two drugs causes inactivation of amikacin. Clinical studies have shown amikacin to be effective in treating bacteraemia, septicaemia including neonatal sepsis and serious infections of the respiratory tract, bones and joints, central nervous system, skin and skin structures (including those resulting from burns), intraabdominal organs, post-operative infections and complicated and recurrent urinary tract infections, when caused by susceptible organisms.

RECOMMENDED DOSAGE

Pharmaniaga Amikacin 500mg/2ml Injection is for use in one patient on one occasion only. Discard any residue.

Uncomplicated infections due to sensitive organisms should respond to treatment within 24 to 48 hours at the recommended dosage. If no improvement occurs within three to five days, the use of amikacin sulfate should be re-evaluated and consideration be given to alternative therapy.

Failure of the infection to respond may be due to resistance of the organism or to the presence of septic foci requiring surgical drainage.

Whenever possible, and especially in patients with impaired renal function, peak and trough amikacin serum concentrations should be determined and dosage adjusted where necessary to maintain desired serum concentrations. In general, desired peak concentrations are between 15 to 30 micrograms/mL, and trough concentrations should not exceed 5 to 10 micrograms/mL. An increased risk of toxicity may be associated with prolonged peak amikacin serum concentrations greater than 30 to 35 micrograms/mL.

Intramuscular or intravenous administration

The intramuscular route is preferred for most infections, but in life-threatening infections or when an intramuscular injection is not feasible, an intravenous infusion (0.25% over 30 to 60 minutes) may be used.

Dosage

Dosage of amikacin sulfate is expressed in terms of amikacin and calculated on a body weight basis. Dosage is identical for both routes of administration.

Adults and children:

The usual recommended dose of amikacin is 15 mg/kg daily given in two or three equally divided doses.

Neonates and premature infants:

Dosage given for patients with normal renal function. Initiate treatment with a loading dose of 10 mg/kg followed by 7.5 mg/kg every 12 hours. The maximum total daily dose should not exceed 15 mg/kg. Solution infusions via the IV route should be given over a 1 to 2 hour period. Insufficient clinical use has not enabled firm dosage guidelines to be established for the use of amikacin in premature infants.

Elderly:

Amikacin is excreted by the renal route. Since renal function could be failing in the elderly, it should be assessed whenever possible and the dosage adjusted accordingly, if necessary.

Urinary tract infections (other than pseudomonal infections):

500 mg/day in two equally divided doses is recommended.

Impaired renal function:

In patients with impaired renal function, either the dose or the dosage interval needs to be adjusted to avoid accumulation. The dosage interval may be calculated using the following formula:

Serum creatinine concentration (mmol /L) x 101.8 = dosage interval (hours)

This formula should not be used to calculate the dosage interval for elderly patients.

Instead, it is advisable to base dosage on creatinine clearance

Serum Creatinine Concentration (mmol / L)		Interval between doses of 7.5 mg/kg (hours)
0.12	x 101.8 =	12
0.15		15
0.20		20
0.25		25.5
0.30		30.5
0.35		35.5
0.40		40.5
0.45		46
0.50		51
0.55		56
0.60		61

As renal function may alter appreciably during therapy, the serum creatinine should be checked frequently and the dosage regimen modified accordingly.

If it is desired to administer amikacin at a fixed time interval, the dosage must be reduced. It is recommended that the drug be given every 12 hours. Serum concentrations of the drug should be measured in these patients to ensure accurate administration and to avoid toxic serum levels.

If serum assay determinations are not available and the patients condition is stable, serum creatinine and creatinine clearance values are the most readily available indicators of the degree of renal impairment to use in determining the dosage.

First, begin therapy by administering 7.5 mg/kg, which is half the normal daily dose.

To determine the size of maintenance doses administered every 12 hours, the initial dose should be reduced in proportion to the reduction in the patients creatinine clearance rate (cc):

Maintenance dose every 12 hours =
$$\frac{\text{Observed cc in mL/min}}{\text{normal cc in mL/min}} \times \text{Calculated initial dose in mg}$$

An alternate rough guide for determining reduced dosage at 12 hour intervals (for patients whose steady state serum creatinine values are known) is to divide the normally recommended dose by the patient's serum creatinine.

The above dosage schedules are provided as guides to dosage when the measurement of amikacin serum levels is not feasible and are not intended to be rigid recommendations.

Mode of Administration

For intramuscular (IM) or intravenous (IV) use.

CONTRAINDICATIONS

Amikacin sulphate injection is contraindicated in patients with hypersensitivity to amikacin or to any of the excipients.

A history of hypersensitivity or serious toxic reactions to aminoglycosides may contraindicate the use of any other aminoglycoside because of the known cross-sensitivities of patients to drugs in this class.

Aminoglycosides may impair neuromuscular transmission, and should not be given to patients with myasthenia gravis.

WARNING AND PRECAUTION

Treatment with amikacin for more than 14 days has not been established as being safe. Patients undergoing treatment with parenteral aminoglycosides should be under close clinical observation and evaluation because of the potential ototoxicity and nephrotoxicity associated with their use.

1) Cross allergenicity:

Cross allergenicity among aminoglycosides has been demonstrated. Therefore, caution should be exercised in patients who have shown hypersensitivity to this class of drugs.

2) Nephrotoxicity:

As with other aminoglycosides, amikacin sulphate is potentially nephrotoxic and neurotoxic; therefore, precautions on dosage and adequate hydration should be observed. Renal function should be closely monitored in patients with known or suspected renal impairment (e.g. diminished glomerular filtration), and also in those whose renal function is initially normal but who develop signs of renal dysfunction during therapy.

Amikacin is present in high concentrations in the renal excretory system; therefore, patients should be well hydrated to minimize chemical irritation/damage of the renal tubules. Prior to initiating therapy, kidney function should be assessed by the usual methods and monitored periodically during the course of treatment.

Hydration should be increased upon signs of renal irritation (casts, white or red cells, or albumin). A reduction in dosage may be desirable if other evidence of renal dysfunction occurs such as decreased creatinine clearance, decreased urine specific gravity, increased serum urea, increased serum creatinine or oliguria. If azotaemia or a progressive decrease in urinary output occurs, treatment should be stopped.

3) Ototoxicity:

In patients treated at higher doses or for periods longer than those recommended, ototoxicity, both auditory and vestibular can occur. Patients with renal damage have the highest risk of developing amikacin induced ototoxicity. The risk of hearing loss due to aminoglycosides increases with the degree of exposure to either high peak or high trough serum concentrations. As with other aminoglycosides, on rare occasions changes in renal and eighth cranial nerve function may not become manifest until soon after completion of therapy. Aminoglycoside induced ototoxicity is usually irreversible.

A pre-treatment audiogram should be obtained and repeated during therapy in patients with renal impairment undergoing treatment for seven days or more as well as in other patients being treated for 10 days. If tinnitus or subjective hearing loss develops; or if follow-up audiograms show significant loss of high frequency response, the use of amikacin sulphate therapy should be discontinued immediately. Lost function may not be fully restored.

4) Neurotoxicity:

Muscular paralysis and neuromuscular blockade have been demonstrated in the cat with high doses of amikacin (188mg/kg). The possibility of respiratory paralysis and neuromuscular blockade should be considered when amikacin is administered concomitantly with neuromuscular blocking drugs or anaesthetics. If blockade occurs, calcium salts may reverse this phenomenon. Other manifestations of neurotoxicity may include numbness, skin tingling, muscle twitching and convulsions.

5) Monitoring of serum levels:
Whenever possible, and especially in patients with renal impairment, peak and trough serum concentrations of amikacin should be determined periodically, and dosage should be adjusted, to maintain desired serum concentrations. In general, desirable peak concentrations of amikacin are 15 to 30 mcg/ml, and trough concentrations of the drug should not exceed 5 to 10 mcg/ml. An increased risk of toxicity may be associated with prolonged peak amikacin serum concentrations greater than 30 to 35 mcg/ml.

6) Concurrent use with other antibiotics or potent diuretics:
Since the risk of ototoxicity, irreversible deafness and nephrotoxicity is increased when amikacin is used in conjunction with the systemic or topical use of other potentially neurotoxic and/or nephrotoxic drugs, such therapy should be avoided. This includes concurrent use with potent diuretics and other aminoglycosides. Other factors which may increase the risk of toxicity are dehydration and advancing age.

7) Geriatric:
Because of its toxicity, amikacin should be used with caution in elderly patients only after less toxic alternatives have been considered and/or found ineffective. Elderly patients are more likely to have an age-related decrease in renal function. This may not be evident in the results of routine screening tests such as serum urea or serum creatinine. A creatinine clearance determination may be more useful. Recommended doses should not be exceeded, and the patient's renal function should be carefully monitored during therapy. Geriatric patients may require smaller daily doses of amikacin in accordance with their increased age, decreased renal function, and possibly, decreased weight. In addition, loss of hearing may result even in patients with normal renal function.

8) Paediatric:
The nephrotoxic and ototoxic potential of amikacin in newborn infants is not known. Until more safety reports are available, amikacin should be used in infants only in those specific circumstances where susceptibility testing indicated that other aminoglycosides cannot be used or are otherwise contraindicated, and when the infant can be observed for evidence of toxicity.

9) Patients with muscular disorders:
Aminoglycosides, including amikacin, should be used with caution in patients with muscular disorders, such as myasthenia gravis or parkinsonism, since these drugs may aggravate muscle weakness because of their potential curare-like effect on neuromuscular junction.

10) Superinfection:
If overgrowth of non-susceptible organisms occurs, appropriate therapy should be initiated.

11) Sulfite-sensitivity:
This preparation contains Sodium metabisulfite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulfites.

12) Topical use:
Aminoglycosides are quickly and almost totally absorbed when they are applied topically, except to the urinary bladder, in association with surgical procedures. Irreversible deafness, renal failure and death due to neuromuscular blockade have been reported following irrigation of both small and large surgical fields with an aminoglycoside preparation.

DRUG INTERACTIONS WITH OTHER MEDICAMENTS

Potent diuretics:
If possible, do not give amikacin in conjunction with ethacrynic acid, frusemide or other potent diuretics which may themselves cause ototoxicity. In addition, when administered intravenously, diuretics may enhance aminoglycoside toxicity by altering antibiotic concentrations in serum and tissue.

Other neurotoxic and/or nephrotoxic agents:
If possible, avoid concurrent or sequential use of other

neurotoxic, ototoxic and/or nephrotoxic antibiotics either systemically or topically, including other aminoglycosides, polymyxin B, bacitracin, ciclosporin, tacrolimus, cephaloridine paromomycin, viomycin, colistin, cisplatin, vancomycin, amphotericin B, clindamycin and cephalosporins. This is because of the potential for additive effects. Where this is not possible, monitor closely.

Increased nephrotoxicity has been reported following concomitant parenteral administration of aminoglycoside antibiotics and cephalosporins. Concomitant cephalosporin use may spuriously elevate creatinine serum level determination.

Anaesthetics/neuromuscular blocking agents or medications with neuromuscular blocking activity:
Concurrent use of amikacin with agents with neuromuscular blocking activity e.g. succinylcholine, tubocurarine, decamethonium, halogenated hydrocarbon inhalation anaesthetics, opioid analgesics and massive transfusions with citrated anticoagulated blood, should be carefully monitored; neuromuscular blockade may be enhanced, resulting in skeletal muscle weakness and respiratory depression or paralysis (apnea); caution is recommended when these medications and amikacin are used concurrently during surgery or in the postoperative period, especially if there is a possibility of incomplete reversal of neuromuscular blockade postoperatively; treatment with anticholinesterase agents or calcium salts may help to reverse the blockade.

Penicillins:
Aminoglycosides are inactivated by solutions containing penicillins. This inactivation is brought about by the opening of the beta-lactam ring and combination of the penicillin with an amino group of the aminoglycoside to form a biologically inactive amide. For this reason, amikacin and penicillins should not be combined in intravenous injections/infusions. The inactivation of some aminoglycosides by penicillins has been reported in vivo, especially in patients with renal failure who maintain a higher level of the penicillin for a longer period of time compared to patients with normal renal function. Therefore, when amikacin and penicillins are used together in patients with renal failure, the time of administration of each drug should be staggered so that several hours separate each infusion.

There is an increased risk of hypocalcaemia when aminoglycosides are administered with bisphosphonates.

There is an increased risk of nephrotoxicity and possibly of ototoxicity when aminoglycosides are administered with platinum compounds.

Concomitantly administered thiamine (vitamin B1) may be destroyed by the active sodium bisulfite component of the amikacin sulphate formulation.

The intraperitoneal use of amikacin is not recommended in patients under the influence of anaesthetics or muscle-relaxing drugs (including ether, halothane, d-tubocurarine, succinylcholine and decamethonium) as neuromuscular blockade and consequent respiratory depression may occur.

Indomethacin may increase the plasma concentration of amikacin in neonates.

Incompatibilities

Chemical and physical inactivation and/or incompatibility have been reported for several other antimicrobial agents, in particular beta-lactam antibiotics (e.g. penicillins, cephalosporins), when mixed with amikacin.

Amikacin is also reported to be incompatible with various other medicinal products (i.e. aminophylline, amphotericin, chlorothiazide sodium, dexamethasone sodium phosphate, erythromycin gluceptate, heparin, nitrofurantoin sodium, phenytoin sodium, potassium chloride, tetracyclines, thiopentone sodium, vitamin B & C complex and warfarin sodium). However, reports are contradictory in many cases, and other factors, such as the strength and composition of the vehicles used, may play a role.

Given their potential for incompatibility, amikacin and other aminoglycosides should not generally be mixed with other medicinal products in syringes or infusion solutions nor given through the same intravenous line. When aminoglycosides are given with a beta lactam, they should generally be given at separate sites.

STATEMENT ON USAGE DURING PREGNANCY AND

LACTATION

Pregnancy

There are limited data on use of aminoglycosides in pregnancy. Aminoglycosides can cause fetal harm. Aminoglycosides cross the placenta and there have been reports of total, irreversible, bilateral congenital deafness in children whose mothers received streptomycin during pregnancy. Although side effects on the fetus or newborns have not been reported in pregnant women treated with other aminoglycosides, the potential for harm exists. In reproduction toxicity studies in mice and rats no effects on fertility or fetal toxicity were reported. If amikacin is used during pregnancy or if the patient becomes pregnant while taking this drug, the patient should be apprised of the potential hazard to the fetus. Amikacin should be administered to pregnant women only when clearly needed and under medical supervision.

The safety of amikacin in pregnancy has not yet been established.

Lactation

It is not known whether amikacin is excreted in breast milk. Since the possible harmful effects on the infant is not known, it is recommended that if nursing mothers must be given amikacin, the infants should not be breast fed during therapy.

ADVERSE EFFECTS/UNDESIRABLE EFFECTS

This list is presented by system organ class, MedDRA preferred term, and frequency using the following frequency categories: very common, common, uncommon, rare, very rare and not known (cannot be estimated from the available data).

System Organ Class	Frequency	MedDRA Term
<i>Infections and Infestations</i>	Uncommon	Superinfections or colonisation with resistant bacteria or yeast
<i>Blood and lymphatic system disorders</i>	Rare	Anaemia, eosinophilia
<i>Immune system disorders</i>	Not known	Anaphylactic response (anaphylactic reaction, anaphylactic shock and anaphylactoid reaction), hypersensitivity
<i>Metabolism and nutrition disorders</i>	Rare	Hypomagnesaemia
<i>Nervous system disorders</i>	Not known	Paralysis
	Rare	Tremor, paresthesia, headache, balance disorder
<i>Eye disorders</i>	Rare	Blindness, retinal infarction
<i>Ear and labyrinth Disorders</i>	Rare	Tinnitus, hypoacusis
	Not known	Deafness, deafness neurosensory
<i>Vascular disorders</i>	Rare	Hypotension
<i>Respiratory, thoracic and mediastinal disorders</i>	Not known	Apnoea, bronchospasm
<i>Gastrointestinal disorders</i>	Uncommon	Nausea, vomiting
<i>Skin and subcutaneous tissue disorders</i>	Uncommon	Rash
	Rare	Pruritus, urticaria
<i>Musculoskeletal, connective tissue and bone disorders</i>	Rare	Arthralgia, muscle twitching

<i>Renal and urinary disorders</i>	Not known	Renal failure acute, nephropathy toxic, cells in urine
	Rare	Oliguria, blood creatinine increased, albuminuria, azotemia, red blood cells urine, white blood cells urine
<i>General disorders and administration site conditions</i>	Rare	Pyrexia

All aminoglycosides have the potential to induce ototoxicity, renal toxicity, and neuromuscular blockade. These toxicities occur more frequently in patients with renal impairment, in patients treated with other ototoxic or nephrotoxic drugs, and in patients treated for longer periods and/or with higher doses than recommended.

Renal function changes are usually reversible when the drug is discontinued.

Toxic effects on the eighth cranial nerve can result in hearing loss, loss of balance, or both. Amikacin primarily affects auditory function. Cochlear damage includes high frequency deafness and usually occurs before clinical hearing loss can be detected by audiometric testing.

Macular infarction sometimes leading to permanent loss of vision has been reported following intravitreal administration (injection into the eye) of amikacin.

When the recommended precautions and dosages are followed the incidence of toxic reactions, such as tinnitus, vertigo, and partial reversible deafness, skin rash, drug fever, headache, paraesthesia, nausea and vomiting is low. Urinary signs of renal irritation (albumin, casts, and red or white cells), azotaemia and oliguria have been reported although they are rare.

OVERDOSAGE

Symptoms may include nephrotoxicity and ototoxicity including disturbances of vestibular function. In severe cases there may be respiratory paralysis due to neuromuscular blockade. Treatment is supportive, as required, and includes ascertaining adequate diuresis. In the event of overdosage or a toxic reaction, peritoneal dialysis or haemodialysis will aid in the removal of amikacin from the blood. Amikacin levels are also reduced during continuous arteriovenous hemofiltration. In the newborn infant, exchange transfusion may also be considered.

STORAGE CONDITION

Store below 30°C.

Do not expose to excessive heat.

The solution will remain stable up to 48 hours at room temperature (<30°C) after mixing with the compatible solutions. Discard any unused portion.

SHELF-LIFE

Products should not be used beyond the expiry date imprinted on the product packaging. Once opened discard any unused portion. The solution will remain stable up to 48 hours at room temperature (<30°C) after mixing with the compatible solutions. Discard any unused portion.

DOSAGE FORMS AND PACKAGING AVAILABLE

• 10 x 2mL vial (clear)

PRODUCT REGISTRATION HOLDER /MANUFACTURER

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